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(a)	Random: Impossible to predict when a particular nucleus in a sample is going to decay.
	Spontaneous: The decay of a nucleus is not affected by by the presence of other nuclei, chemical reactions or physical conditions such as temperature and pressure.
(b)(i)	${}_{84}^{210}\text{Po} \rightarrow {}_{82}^{206}\text{Pb} + {}_2^4\text{He}$

(b)(ii)	$KE = \frac{p^2}{2m}$ Since magnitude of p is the same, $\frac{KE_{\alpha}}{KE_{Pb}} = \frac{m_{Pb}}{m_{\alpha}} = \frac{206}{4}$
	$\frac{KE_{\alpha}}{KE_{Pb}} = 51.5$
(b)(iii)	$A_0 = \lambda N_0 = \frac{\ln 2}{138 \times 24 \times (60)^2} (4.2 \times 10^{11}) = 2.44 \times 10^4 \text{ Bq}$
	$A = A_0 e^{-\left(\frac{\ln 2}{t_{1/2}}\right)t} = A_0 e^{-\left(\frac{\ln 2}{138}\right)600}$
	$A = 1200 \text{ Bq (2 s.f.)}$
(b)(iv)	The reading will be lower. Particles are emitted in all directions and only a proportion will be detected by the counter.
(c)(i)	By conservation of momentum, for a two particle system, the ratio of the speeds of the particles is fixed (inverse ratio of masses).
	By conservation of energy, energy released per decay is constant, the speed of the beta particle emitted should be constant. However, the range of KE suggest that there must be a third particle.
(c)(ii)	Half-life of platinum is much longer than that of gold.
	Formation of gold is slower than its decay, hence gold will be the nuclei of the smallest percentage.
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